

Q-Shield: A Quantum-Inspired Behavioral Biometric Authentication Framework with Post-Quantum Cryptographic Protection

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Abstract

Traditional authentication mechanisms — including passwords, one-time tokens, and static biometrics — remain critically vulnerable to phishing, credential stuffing, session hijacking, and adversarial spoofing. Compounding these threats, the projected advent of cryptographically relevant quantum computers renders widely deployed classical cryptographic primitives such as RSA and ECC fundamentally insecure, while existing post-quantum biometric systems address only template-level cryptographic protection without modifying the underlying biometric representation itself. Simultaneously, behavioral biometric systems capable of continuous session monitoring apply no quantum-resistant security layer and rely exclusively on time-domain feature representations with classical similarity metrics — leaving both the identity signal and its cryptographic envelope vulnerable. To close this gap, we propose Q-Shield: a quantum-inspired secure authentication framework that uniquely unifies six capabilities absent in any single prior system. Q-Shield captures multi-modal behavioral signals — keystroke dynamics, mouse movement patterns, and touch interaction timing — and transforms them via Fast Fourier Transform (FFT) into normalized spectral signatures interpreted as amplitude-encoded quantum-style identity state vectors. Authentication is performed using quantum fidelity as the similarity metric, providing a richer and more spoofing-resistant matching mechanism than conventional Euclidean or cosine distance measures. Identity templates are secured end-to-end using NIST-standardized post-quantum cryptographic algorithms — ML-KEM (CRYSTALS-Kyber) for key encapsulation and ML-DSA (CRYSTALS-Dilithium) for digital signatures — with an optional Quantum Key Distribution channel for the highest-security deployment contexts. A machine learning anomaly detection engine performs continuous, session-long behavioral monitoring, triggering re-authentication or session revocation upon detecting statistically significant deviations from the enrolled identity profile. Unlike prior work that either protects static biometric templates cryptographically or applies quantum-inspired modeling without security integration, Q-Shield makes the behavioral representation itself inherently harder to replicate while simultaneously ensuring quantum-resistant storage, transmission, and verification of identity data. The framework targets high-security domains including defense systems, critical infrastructure, healthcare information systems, and financial services platforms where both present-day adversarial threats and future quantum decryption risks must be jointly addressed.

Behavioral Biometrics, Post-Quantum Cryptography, Continuous Authentication, Quantum-Inspired Computing, FFT Spectral Encoding, Quantum Fidelity, Keystroke Dynamics, ML-KEM, ML-DSA, Anomaly Detection, Session Security, Lattice-Based Cryptography

1 Literature Survey

This section synthesizes the 30 surveyed papers into thematic categories, tracing the evolution of each research thread and explicitly identifying how Q-Shield extends, integrates, and advances beyond each cluster of prior work.

1.1 Behavioral Biometrics: Surveys and Multimodal Frameworks

The field of behavioral biometrics has matured significantly since early keystroke studies, with comprehensive surveys now covering nearly every passive interaction modality. Stylios et al. [?] provide the canonical survey of behavioral signals for mobile continuous authentication, classifying keystroke dynamics, mouse movements, touch gestures, gait, and device handling patterns. Their work identifies adversarial spoofing and behavioral drift as the two most pressing open challenges—both of which Q-Shield directly addresses. Liang et al. [?] extend this landscape to the IoT era, arguing that AI-driven behavioral analytics can enable passive, continuous identity assurance on resource-constrained devices. The most recent comprehensive review, Ayeswarya and Singh [?] (2024), surveys the intersection of biometric profiling and continuous authentication, cataloguing deep learning advances through 2023. Critically, all three surveys treat quantum computing as a purely cryptographic threat rather than a modeling opportunity—a gap Q-Shield explicitly closes by leveraging quantum-inspired spectral representations of behavioral signals.

1.2 Keystroke Dynamics

Keystroke dynamics remain the most extensively studied behavioral modality. Shadman et al. [?] provide the definitive 2023 survey covering hold time, flight time, and n-graph features, along with comparative ML model analysis across public datasets. Meenakshisundaram et al. [?] demonstrate that combining CNN and LSTM architectures for keystroke sequence modeling achieves lower equal error rates than classical baselines. Chang, Li, and Stamp [?] establish important fixed-text and free-text baselines, finding XGBoost and MLP competitive in fixed-text settings while deep sequence models excel for free-text scenarios. Q-Shield builds directly on these feature engineering insights—specifically adopting hold time and flight time sequences as the raw input to its FFT spectral encoding pipeline—while extending beyond these works by transforming time-domain keystroke features into a frequency-domain identity state vector for quantum-inspired matching.

1.3 Mouse and Pointer Dynamics

Mouse dynamics provide a complementary and equally discriminative behavioral signal. Chong and Elovici [?] demonstrate that CNN architectures with transfer learning outperform classical ML on small mouse dynamics datasets, while Ciaramella et al. [?] show that explainable deep learning (SHAP-augmented models) can identify which trajectory features are most user-discriminative on a continuous basis. These works validate the use of mouse velocity, angle, and curvature as persistent identity markers. Q-Shield incorporates mouse dynamics as a second channel into its FFT-based identity state vector, enabling multi-signal spectral fusion that neither of these works explores.

1.4 Touch and Mobile Interaction Dynamics

Touch dynamics have emerged as a critical modality given the proliferation of smartphones. Qiao et al. [?] introduce TouchSeqNet, a Transformer-based contrastive learning system achieving strong cross-dataset generalization for swipe and tap authentication. DeRidder et al. [?] present a multi-finger touch dynamics dataset from mobile games achieving 93% accuracy with Random Forest classifiers. Qatibani et al. [?] propose a TOPSIS-based hybrid ML framework incorporating touch pressure, trajectory, and velocity features. Together, these works confirm that touch patterns are highly distinctive and suitable for passive continuous authentication—motivating Q-Shield’s touch/tap interaction channel within its behavioral capture layer.

1.5 Privacy-Preserving and Continuous Authentication

Privacy-preserving continuous authentication is a critical system requirement. Baig, Eskeland, and Yang [?] propose homomorphic encryption with oblivious transfer protocols for continuous authentication using keystroke and swipe data, demonstrating that biometric features can be compared on an authentication server without revealing raw signals. Hu et al.’s Auth-ConFormer [?] introduces a convolutional Transformer for sensor-based continuous smartphone authentication, achieving state-of-the-art performance. Neither work, however, employs post-quantum standardized cryptographic primitives (CRYSTALS-Kyber/Dilithium), leaving template storage and transmission vulnerable to future quantum attacks. Q-Shield addresses this gap directly.

1.6 Post-Quantum Biometric Template Protection

The protection of biometric templates against quantum-capable adversaries has received growing attention. Arjona et al. [?] demonstrate that Classic McEliece homomorphic encryption can protect FaceNet facial embeddings with over 90% storage reduction and real-time performance. Palma and Montessoro [?] extend PQC template protection to multimodal (face, fingerprint, iris) biometrics using CRYSTALS-Kyber/Dilithium, showing negligible accuracy degradation. Kuznetsov et al. [?] further derive cryptographic keys directly from CNN facial features under code-based PQC schemes. Critically, all these systems protect biometric templates at the cryptographic layer without modifying the biometric representation itself—meaning the underlying feature vectors remain as vulnerable to adversarial mimicry and reconstruction as in classical systems. Q-Shield’s FFT spectral encoding makes the representation itself harder to replicate.

1.7 Post-Quantum Continuous Authentication for Healthcare

The most closely related prior work to Q-Shield is Bera et al. [?], who propose a post-quantum continuous authentication framework for healthcare using behavioral biometrics and vector similarity search, specifically designed to replace RSA/ECC-based one-time login with a quantum-resistant session-long mechanism. Their work validates both the healthcare domain importance and the combination of behavioral biometrics with PQC. However, their system uses standard Euclidean or cosine vector similarity rather than quantum fidelity as a matching metric, and does not encode behavioral signals in the frequency domain as amplitude-encoded quantum state vectors. Q-Shield extends this paradigm by introducing both FFT-based spectral encoding and quantum fidelity matching.

1.8 Quantum-Inspired Modeling for Authentication and Security

Quantum-inspired computation has recently entered the biometric authentication and cybersecurity space. Dahan and Jaha [?] introduce IRIS-QResNet, demonstrating that quantum-inspired layers improve iris recognition accuracy by up to 16.66%, validating the benefit of quantum-inspired architectural elements for biometric discrimination. Reddy et al. [?] apply quantum kernel methods and amplitude encoding to cyber threat classification, demonstrating that quantum-inspired feature representations can outperform classical baselines on cybersecurity datasets. These works establish the foundational principle that quantum-inspired representations improve discriminability—but neither applies this insight to behavioral biometrics or combines it with post-quantum cryptographic template protection, which Q-Shield uniquely achieves.

1.9 PQC Standards, Cryptographic Infrastructure, and IoT

The standardization landscape for post-quantum cryptography was formalized in August 2024 when NIST released FIPS 203 (ML-KEM/Kyber), FIPS 204 (ML-DSA/Dilithium), and FIPS 205 (SLH-DSA/SPHINCS+) [?]. Joseph et al. [?] provide the organizational transition roadmap underpinning Q-Shield’s threat model, particularly the harvest-now-decrypt-later attack vector. Dobias et al. [?] demonstrate that unified FPGA implementations of Kyber and Dilithium are resource-efficient enough for edge and IoT deployments, supporting Q-Shield’s feasibility on embedded authentication nodes. In the IoT and healthcare domain, Mansoor et al. [?] and the 2026 Scientific Reports protocol [?] confirm that Kyber-512/Dilithium-2 are appropriate for resource-constrained deployments while Kyber-1024/Dilithium-5 serve higher-security gateway roles.

1.10 Hybrid QKD-PQC Transport Security

For the most sensitive deployment contexts, Q-Shield incorporates an optional Quantum Key Distribution transport layer. Garms et al. [?] experimentally demonstrate a modular hybrid QKD+PQC cryptosystem on an FPGA prototype, showing that QKD (information-theoretically secure) and PQC (computationally secure and scalable) provide complementary and synergistic protection. Tian et al. [?] address the finite-size key challenge in hybrid QKD+PQC authentication for constrained channels, developing privacy amplification optimizations that maintain composable security under limited key budgets. These works directly inform Q-Shield’s optional QKD channel design.

1.11 Anomaly Detection and Session Monitoring

Q-Shield’s continuous session monitoring layer draws on unsupervised anomaly detection research. Sharma and Pokharel [?] demonstrate that LSTM Autoencoders can model normal user session behavior and detect anomalies via reconstruction error with 90.17% accuracy in insider threat scenarios—establishing the ML architecture pattern for Q-Shield’s decision engine. Gujjala and Gudavalli [?] demonstrate that heuristic-optimized multi-model biometric fusion further improves anomaly detection robustness. Together these establish that unsupervised deep learning over behavioral sequences is mature enough for real-time session monitoring.

1.12 Multi-Factor and Financial Authentication

In the financial sector, Gujjala P. [?] proposes the most comprehensive prior quantum-enhanced multi-factor authentication framework for banking, integrating lattice-based PQC with behav-

ioral analytics and deep learning classifiers. Their work motivates Q-Shield’s financial services use case and demonstrates industrial viability. Elmisery et al. [?] analyze the dual threat landscape where AI-powered adversarial attacks combine with quantum computing threats against financial transaction security, providing strong justification for Q-Shield’s defense-in-depth architecture that addresses both threats simultaneously.

The table below provides a direct feature-by-feature comparison of Q-Shield against the most representative existing systems. Seven capability dimensions are evaluated: (1) Behavioral Biometric Capture, (2) FFT-Based Spectral Encoding of behavioral features, (3) Quantum Fidelity as a similarity metric, (4) Post-Quantum Cryptographic Protection of identity templates, (5) Continuous Session Authentication beyond one-time login, (6) ML-Based Anomaly Detection for real-time session monitoring, and (7) Optional QKD Transport Layer for highest-security deployments.

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Table 1: Research Gap Analysis and Feature Comparison

Reference	BC	FFT	QF	PQC	CA	AD
Stylios et al. [6] (2021)	Y	N	N	N	Y	N
Liang et al. [12] (2020)	Y	N	N	N	Y	N
Shadman et al. (2023)	Y	N	N	N	N	N
Baig et al. (2023)	Y	N	N	P	Y	N
Arjona et al. [9] (2023)	N	N	N	Y	N	N
Palma & Montessoro [10] (2025)	N	N	N	Y	N	N
Bera et al. [7] (2025)	Y	N	N	Y	Y	N
Dahan & Jaha (2025)	N	N	P	N	N	N
Reddy et al. (2024)	N	Y	P	N	N	N
Kuznetsov et al. (2023)	N	N	N	Y	N	N
Garms et al. (2024)	N	N	N	Y	N	N
Gujjala P. [11] (2023)	Y	N	N	Y	N	N
Sharma & Pokharel (2020)	N	N	N	N	Y	Y
Elmisery et al. [2] (2025)	N	N	N	N	N	N
Q-Shield (Proposed)	Y	Y	Y	Y	Y	Y

BC: Behavioral Capture, **FFT**: FFT Spectral Encoding, **QF**: Quantum Fidelity Matching, **PQC**: Post-Quantum Cryptographic Protection, **CA**: Continuous Authentication, **AD**: ML-based Anomaly Detection. Y = Yes, N = No, P = Partial.

As shown in Table 1, existing studies typically address only a subset of the capabilities required for next-generation authentication systems. Behavioral biometric works focus on data acquisition and continuous authentication but lack quantum-resistant security mechanisms. PQC-based approaches protect biometric templates but do not enhance the underlying behavioral representation. Quantum-inspired authentication studies improve feature discriminability but omit continuous monitoring and cryptographic protection. In contrast, Q-Shield uniquely integrates multi-modal behavioral capture, FFT-based spectral encoding, quantum fidelity matching, NIST-standardized post-quantum cryptography, and machine-learning-based anomaly detection within a unified continuous authentication framework.

2 Use Cases

The framework targets high security, high impact domains:

1. Defense and Intelligence Systems: Continuous, quantum resistant access control for classified systems and command and control infrastructure where session hijacking or credential theft could have strategic consequences.
2. Critical Infrastructure Control: Protection for control consoles managing power grids, water systems, and transportation networks, ensuring only authenticated operators remain in control throughout a session.
3. Healthcare Information Systems: Securing EHR and clinical workstations with continuous monitoring to prevent session hijacking in hospitals and clinics, while resisting future quantum decryption threats to stored identity data.
4. Financial Services and Trading Platforms: Quantum resistant authentication for online banking, payment systems, and high frequency trading platforms, where behavioral biometrics plus Post Quantum encryption reduce fraud and “harvest-now, decrypt-later” risks

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